

Worksheet W14: Epilogue — The Horizon

"You've learned the language of change. Now it's time to see it everywhere."

Epilogue · 12 Problems · Name: _____ Date: _____

The Horizon — Where the known meets the unknown.

Problem E1

Question: What are the **two big questions** of calculus?

💡 *Hint: Speedometer and odometer.*

Problem E2

Question: Why is **optimization** important in **data science**?

💡 *Hint: Think about training models and minimizing error.*

Problem E3

Question: How does calculus help in **physics**?

💡 *Hint: Think about motion, speed, acceleration.*

Problem E4

Question: What role does e^x play in **finance and economics**?

💡 *Hint: Think about continuous compounding interest.*

Problem E5

Question: How are **sine and cosine** used in **medicine**?

💡 *Hint: Think about heart rhythms and EKGs.*

Problem E6

Question: How does calculus help **AI and machine learning** learn?

💡 *Hint: Think about gradient descent and optimization.*

Problem E7

Question: The **final code** of calculus: What does it tell you?

💡 *Hint: Think about the derivative, integral, and Fundamental Theorem.*

Problem E8

Question: What does the **Mental Zoom** show you about the world?

💡 *Hint: Think about seeing patterns everywhere.*

Problem E9

Question: What are the **next steps** after completing this book?

💡 *Hint: Practice, go deeper, apply, teach.*

Problem E10

Question: In one sentence, what is **calculus**?

💡 *Hint: Think about arithmetic and motion.*

Problem E11

Question: Why does the **horizon** represent the beginning, not the end?

💡 *Hint: Think about the journey of learning.*

Problem E12

Question: What is the **final promise** of this book?

💡 *Hint: Think about who can learn calculus.*